Jackfruit Phase-1

Software Requirements Specification

for

Web Based ML Model Evaluator

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PESU

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Revision History

Name	Date	Reason For Changes	Version

Introduction

Purpose

This document's purpose is to demonstrate detailed specifications of the Web Based ML Model Evaluator. Please find the defined functional capabilities, external interface behaviors, system architecture constraints and non functional requirements for the platform. This will serve as a reference to develop, test and verify our project throughout the process.

Intended Audience

This document is intended for the following readers:

- Developers: To implement the backend and UI of the project.
- Testers: To understand the work flow, required inputs, expected outputs and build test cases from the details.
- Evaluator: To review our work.
- Users: The ones who will use the software.

Product Scope

This software is designed to help users to figure out how different types of machine learning models and algorithms work for a selected problem using metrics that help evaluate the most model/algorithm and come up to a conclusion.

Below are some key system capabilities:

- Data ingestion: allows user to upload the dataset that they want to work on.
- Data pre-processing: clean the data before training and train/test split of data.
- Task routing: The software should support various different types of machine learning problem types like classification, regression and clustering.
- Model execution: applying core algorithms like logistic regression, decision trees, KNN, linear regression, etc.
- Performance evaluation: Generate performance metrics such as accuracy, F1 Score, etc which will help the user determine and select the best model and algorithm.
- Visualization: Generate visual outputs like confusion matrix, 2D PCA scatter plots, etc for better readability and understanding.
- Side by side metric comparison enabling users to compare performance between multiple trained models simultaneously.

References

- IEEE Recommended Practice for Software Requirements Specifications.
- Documents created by PESU and slides.

Overall Description

2.1 Product Perspective

The Web Based ML Model Evaluator is a self-contained web application designed to streamline the machine learning workflow. It provides an end-to-end pipeline from data ingestion to model evaluation within a unified browser interface.

2.2 Product Functions

- Dataset uploading and CSV preview generation.
- Data preprocessing, including missing value handling.
- ML task routing for Classification, Regression, and Clustering.
- Feature/target selection and train/test partition adjustments.
- Task-specific algorithm execution across multiple standard models.
- Performance metric computation and visual model comparison.

2.3 User Classes and Characteristics

The primary users are individuals seeking an intuitive, guided process to evaluate machine learning models. Users require a clear workflow that progresses sequentially from dataset upload to result visualization.

2.4 Operating Environment

The application operates entirely within modern web browsers. It must run smoothly on Google Chrome, Firefox, and Microsoft Edge. The system functions independently without requiring any additional browser plugins.

2.5 Design and Implementation Constraints

The system enforces a maximum file upload limit to prevent server performance degradation. Processing times for ingestion, feature scaling, model fitting, and metric generation are strictly constrained to a maximum of 30 seconds for datasets under 10 MB.

2.6 Assumptions and Dependencies

The system assumes that user data is provided specifically in a CSV file format. It depends on a backend architecture capable of intercepting invalid execution states to maintain stability.

External Interface Requirements

3.1 User Interfaces

The user interface provides a step-by-step guided experience: upload dataset, select columns, train model, and display results. The UI displays a detailed dataset summary view showing total rows, column names, data types, missing value counts, and a 5-row preview. It features interactive configuration modules for task selection, cluster inputs, and splitting ratios. Evaluation screens include a model comparison dashboard, performance metrics, and visuals such as 2D Scatter Plots generated via PCA. The UI is also responsible for presenting human-readable error messages.

3.2 Software Interfaces

The system requires a robust backend software interface for data validation and error handling. This backend must capture invalid execution states and transmit structured error data back to the user interface. The software relies on CSV parsing modules to analyze uploaded datasets.

3.3 Communication Interfaces

The application relies on standard browser communication protocols to function seamlessly across Chrome, Firefox, and Edge. Client-server communications must be optimized to ensure that the entire pipeline response latency remains under the 30-second threshold for permitted file sizes.

Analysis Models

Use case diagram:

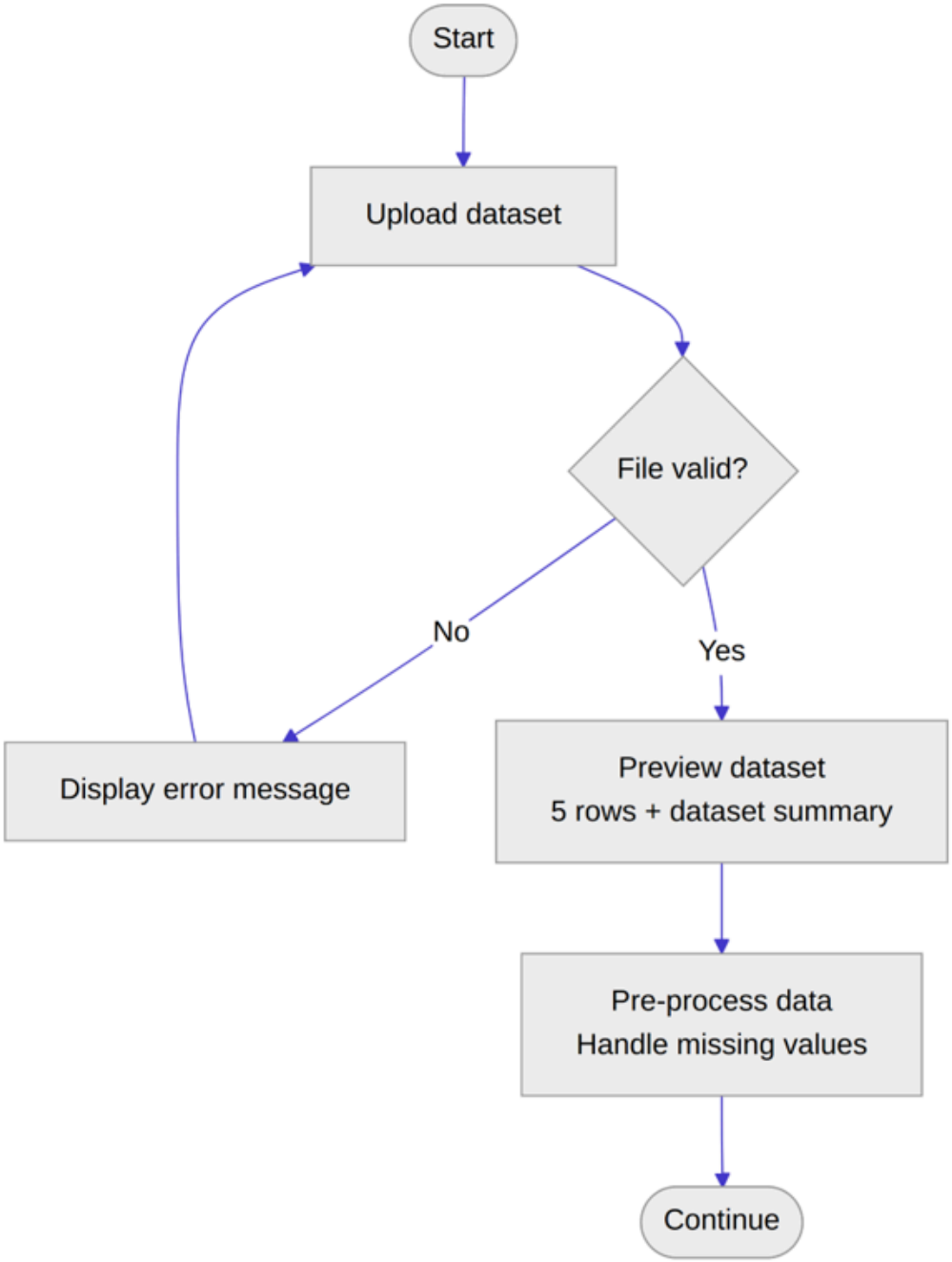


Analysis Model 1: Dataset Ingestion and Preprocessing

Covers:

- Upload dataset
- 5-record preview
- Pre-process data

Activity diagram:

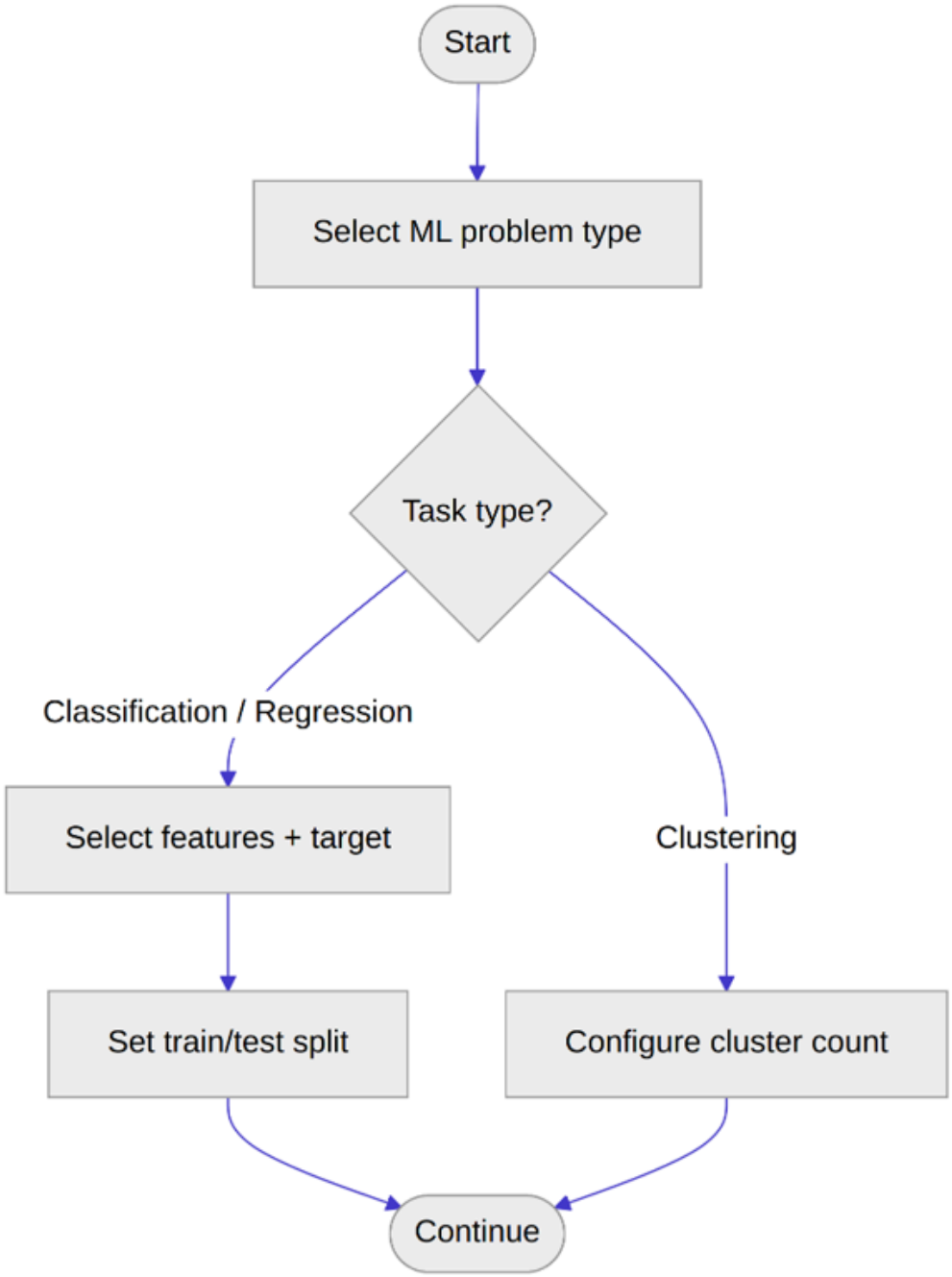


Analysis Model 2: ML Task Configuration

Covers:

- Select ML problem type
- Select features + target (Classification / Regression)
- Configure clustering parameters (Clustering)
- Set train-test split

Activity Diagram:

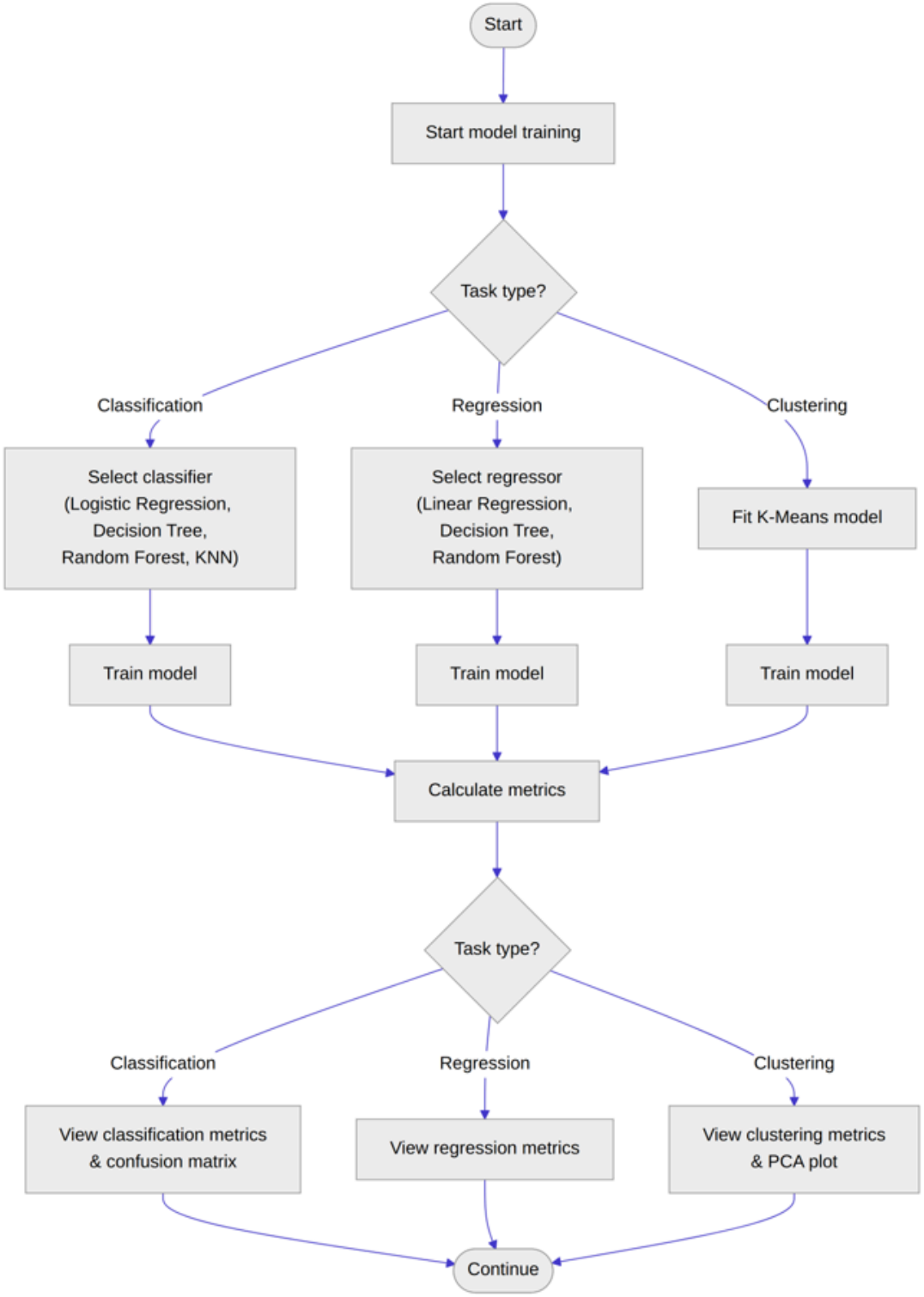


Analysis Model 3: Model Training & Evaluation

Covers:

- Start model training
- Select algorithm (Classifier / Regressor / K-Means) and train
- Calculate metrics
- Classification / Regression / Clustering metric views
- Confusion matrix
- PCA plot

Activity Diagram:



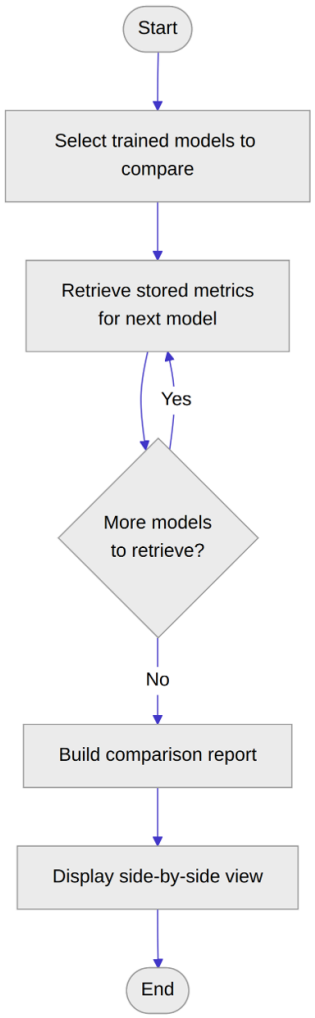
Analysis Model 4: Model Comparison

Covers:

- Trained Models
- Collect evaluation metrics
- Compare models
- Display side-by-side

Activity Diagram

Shows the end-to-end flow: selecting models, looping through each to retrieve its stored metrics, then building and displaying the comparison.



System Features

System Feature 1: Data Ingestion and Preprocessing (5.1)

5.1.1 Description and Priority

Allows users to upload dataset in csv format, preview 5 records from dataset, and handle basic missing values and data pre processing.

Priority: High

Risk involved in uploading dataset is comparatively low whereas risk involved in performing pre processing steps on the uploaded dataset can be challenging since all the datasets have variation and hence the risk involved in this is high. We could involve having certain different types of options for handling missing values such as mean/median/drop rows etc.

5.1.2 Stimulus/Response Sequences

- **User action:** Selects file and clicks upload.
- **System response:** Checks file size, file type, reads the file and previews first 5 records from the uploaded dataset, also performs pre processing and then displays results.

5.1.3 Functional Requirements

REQ-1: The system should allow users to upload CSV files upto a maximum size of 10 MB
REQ-2: The system should display dataset summary of first five records.
REQ-3: The system should provide pre processing options to clean the dataset by handling missing values.

System Feature 2: ML Task Routing and Configuration (5.2)

5.2.1 Description and Priority

Allows users to specify the ML problem type, assign feature and target columns, configure other algorithm parameters, split train/test data.

Priority: High

ML model performance will depend on these inputs from user but implementing this involves low risk.

5.2.2 Stimulus/Response Sequences

- **User Action:** Chooses an ML problem type like classification/regression/clustering.

- **System Response:** Renders relevant column selection drop downs and hyperparameter inputs based on the chosen task.
- **User Action:** Selects target/feature variables and adjusts the train/test split ratio (or specifies the number of clusters for unsupervised tasks).
- **System Response:** Validates selections to prevent invalid states (e.g., selecting the same column as both feature and target) and stores configuration parameters.

5.2.3 Functional Requirements

REQ-4: The system should allow users to select an ML task type like classification, regression or clustering before configuring model parameters.

REQ-5: For supervised tasks like classification and regression, the system should allow users to select one column as the target variable and one or more columns as feature variables.

REQ-6: For clustering tasks, the system should allow users to select feature columns and specify the target number of clusters.

REQ-7: For supervised tasks, the system should allow user to input the train/test data partition ratio.

System Feature 3: Algorithm Execution and Metric Computation (5.3)

5.3.1 Description and Priority

Executes selected machine learning algorithms against the prepared dataset, calculates performance metrics, and generates visualization charts.

Priority: High

This section is very crucial to the software and since, risk involved can be high.

5.3.2 Stimulus/Response Sequences

- **User Action:** Selects target algorithms and clicks execute.
- **System Response:** Fits the selected models, evaluates performance on test/clustered data for files under 10 MB and generates evaluation metrics and charts.

5.3.3 Functional Requirements

REQ-8: The system should execute selected machine learning models based on user choice- logistic regression, decision tree, random forest, KNN for classification problems whereas for regression problems, linear regression, decision tree regressor and random forest regressor and K means for Clustering.

REQ-9: For classification models, the system should calculate and display Accuracy, Precision, Recall, F1-Score and a Confusion Matrix.

REQ-10: For regression models, the system should calculate and display mean absolute error, mean squared error (MSE), root mean squared error.

REQ-11: For clustering models, the system should calculate silhouette score and Davies-Bouldin Index, and generate a 2D PCA scatter plot showing cluster assignments.

System Feature 4: Model Comparison Dashboard and Error Handling (5.4)

5.4.1 Description and Priority

Provides a side by side interface comparing metrics across multiple trained models and intercepts execution errors safely.

Priority: Medium.

Visualization can be easily done so the risk is low, but for error handling the risk involved is high because we may encounter various types of errors and we should be able to deal with all of them.

5.4.2 Stimulus/Response Sequences

- **User Action:** Selects multiple trained model runs to compare.
- **System Response:** Displays a consolidated side by side metric comparison dashboard.
- **User Action:** Submits an invalid dataset or configuration.
- **System Response:** Software throws an error.

5.4.3 Functional Requirements

REQ-12: The system should provide a comparison dashboard enabling users to view evaluation metrics of two or more trained models side by side.

REQ-13: The backend should intercept invalid execution states or invalid data formats and return error message to the UI.

Other Nonfunctional Requirements

6.1 Performance Requirements

The following performance requirements define the expected execution time and resource constraints of the Web-Based ML Model Evaluator.

- **PR-1: Pipeline Response Time**

The system shall complete dataset ingestion, feature scaling, model fitting, and metric generation within 30 seconds per run for datasets smaller than 10 MB.

- **PR-2: File Upload Constraint**

The system shall restrict the maximum size of uploaded dataset files to prevent excessive server load and performance degradation.

6.2 Safety Requirements

The following safety requirements define safeguards to prevent invalid or potentially harmful system operations during dataset processing and model evaluation.

- **SR-1: Safe Data Processing**

The system shall validate uploaded datasets and input configurations before processing to prevent invalid data from causing failures during preprocessing or model execution.

- **SR-2: Safe Model Execution**

The system shall prevent model execution when the selected task, features, target, or required parameters are invalid or incomplete.

- **SR-3: Error Handling**

The system shall handle invalid execution states without causing system failure and shall provide structured, human-readable error messages to the user interface.

6.3 Security Requirements

The following security requirements define measures for protecting user-provided datasets and preventing unauthorized or unsafe inputs from affecting the system.

- **SEC-1: Dataset Protection**

The system shall protect uploaded datasets from unauthorized access or disclosure during processing.

- **SEC-2: Input Validation**

The system shall validate uploaded files and user-provided inputs before processing to prevent malicious or invalid input from being executed by the system.

- **SEC-3: Secure Data Handling**

The system shall ensure that uploaded datasets and generated evaluation results are handled securely and shall not be exposed to unauthorized users.

- **SEC-4: Backend Security**

The backend shall reject requests containing invalid or unsupported inputs and shall return appropriate error messages without exposing internal system information.

6.4 Software Quality Attributes

The following quality attributes define the expected overall quality and usability of the Web-Based ML Model Evaluator.

- **QA-1: Usability**

The system shall provide a clear and intuitive workflow that guides users through the process of **dataset upload -> column selection -> model training -> result display**.

- **QA-2: Reliability**

The system shall handle invalid datasets, configurations, and execution states without causing system failure and shall provide appropriate feedback to the user.

- **QA-3: Portability**

The web application shall operate on modern web browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge, without requiring additional browser plugins.

- **QA-4: Maintainability**

The system shall be organized into separate functional components for data ingestion and preprocessing, ML task configuration, model execution and evaluation, and model comparison, allowing individual components to be modified or extended without significantly affecting the others.

- **QA-5: Robustness**

The system shall validate user inputs and dataset conditions before performing preprocessing or model execution, and shall gracefully handle invalid inputs and execution errors.

- **QA-6: Testability**

The system shall provide clearly defined inputs and outputs for its major functional components so that dataset processing, model training, evaluation, and result generation can be independently tested.

6.5 Business Rules

The following business rules define the conditions under which the Web-Based ML Model Evaluator performs its major functions.

- **BR-1: Dataset Requirement**

The user shall upload a valid dataset before proceeding with dataset preview, preprocessing, or model training.

- **BR-2: Supervised Learning Configuration**

For Classification and Regression, the user shall select one target variable and one or more feature variables before model training.

- **BR-3: Clustering Configuration**

For Clustering, the user shall specify the number of clusters before K-Means model execution.

- **BR-4: Train/Test Split**

The **train/test split ratio** shall be configurable only for supervised learning tasks such as Classification and Regression.

- **BR-5: Task-Specific Algorithms**

The system shall execute algorithms according to the selected ML problem type:

1. **Classification:** Logistic Regression, Decision Tree Classifier, Random Forest Classifier, and KNN.
2. **Regression:** Linear Regression, Decision Tree Regressor, and Random Forest Regressor.
3. **Clustering:** K-Means Clustering.

- **BR-6: Task-Specific Evaluation**

The system shall display evaluation metrics appropriate to the selected ML task:

1. **Classification:** Accuracy, Precision, Recall, F1-Score, and confusion matrix.
2. **Regression:** MAE, MSE, RMSE, and R^2.
3. **Clustering:** Silhouette Score, Davies-Bouldin Index, and PCA-based visualization.

- **BR-7: Model Comparison**

Model comparison shall be performed only for trained models with available evaluation results, and the system shall present the selected models' evaluation metrics side-by-side.

Other Requirements

7.1 System Maintenance & Scalability

- Modular Pipeline Architecture: The core backend must decouple data processing routines using pandas, algorithm fitting pipelines like scikit-learn, and API routes to ensure independent maintenance and allow seamless integration of new ML models in future releases.
- Deprecations & Libraries: All third party Python package dependencies such as scikit-learn, pandas, and numpy) must be locked to specific stable version releases to prevent build breaks from breaking API changes.

7.2 Error Handling & Fault Tolerance

- The execution engine must capture and wrap dataset exceptions (e.g. non numeric data in regression targets, zero-variance columns, or convergence timeouts) without terminating the server process.
- User facing diagnostics: Server side computational errors must be caught by custom backend middleware and returned as structured, human readable error messages to the UI.

7.3 Data Privacy & Memory Management

- Uploaded tabular datasets and generated visualization artifacts must be processed in memory or stored in temporary session buffers.
- Uploaded CSV files and intermediate model objects must be automatically purged from host memory upon session termination to conserve server memory resources.

Appendix A: Glossary

- **SRS:** Software Requirements Specification
- **ML:** Machine Learning
- **CSV:** Comma Separated Values
- **Supervised Learning:** Machine learning tasks (Classification and Regression) where models are trained using labeled target data.
- **Unsupervised Learning:** Machine learning tasks (Clustering) that discover hidden patterns or groupings in unlabeled datasets.
- **PCA:** Principal Component Analysis, a dimensionality reduction technique.
- **MAE:** Mean Absolute Error
- **MSE / RMSE:** Mean Squared Error/Root Mean Squared Error
- **R^2 Score:** Coefficient of Determination
- **Silhouette Score:** A clustering metric measuring how similar an object is to its own cluster compared to other clusters (ranging from -1 to +1).
- **Davies-Bouldin Index:** A clustering evaluation index where lower values signify better separation and tighter cluster groupings.

Appendix B: Field Layouts

Table 1: Dataset Configuration & Preprocessing Inputs

Field	Length	Data Type	Description	Mandatory?
Dataset File	Max 10 MB	CSV file	The user selects CSV dataset file	Yes
ML Task Type	20	Dropdown	Target task: Classification, Regression or Clustering	Yes
Target Variable	60	String/ Dropdown	Selected column name to predict (Supervised tasks only)	Yes (Supervised) / No (Clustering)
Train/Test Split Ratio	4	Float	Percentage of data reserved for testing	Yes (Supervised)

B.2 Output Report Field Layouts

Table 2: Model Performance Evaluation Metrics Report

Metric Field Name	Data Type	Source/Task	Description
Model Name	String	All	Name of the evaluated algorithm

Accuracy Score	Float (0 to 1)	Classification	Ratio of correct predictions over total predictions
Precision Score	Float (0 to 1)	Classification	Ratio of true positives over total positive predictions
Recall Score	Float (0 to 1)	Classification	Ratio of true positives over total actual positives
F1-Score	Float (0 to 1)	Classification	Harmonic mean of precision and recall
Confusion Matrix Data	2D Array/ JSON	Classification	Counts of true vs. predicted class values
Mean Absolute Error (MAE)	Float	Regression	Average absolute difference between predicted and actual values
Mean Squared Error (MSE)	Float	Regression	Average squared difference between predicted and actual values
Root MSE (RMSE)	Float	Regression	Square root of the Mean Squared Error
R2 Score	Float	Regression	Coefficient of determination

Silhouette Score	Float	Clustering	Measure of cluster cohesion and separation
Davies-Bouldin Index	Float	Clustering	Ratio of within-cluster scatter to between-cluster separation

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1.	FR-1: Dataset Upload	User uploads dataset					
2.	FR-2: Dataset preview	5 record preview of uploaded dataset					
3.	FR-3: Preprocessing and Cleaning	Preprocessig and Cleaning of data, removing missing values if any					
4.	FR-4: ML Task Routing	select ML problem type					
5.	FR-8: Task specific Algorithm Execution	Execution of algorithms					
6.	FR-12: Model Comparison Visualizations	View comparison metrics and charts					